

Four Ways My Eval Lied to Me

Building an evaluation harness for contract clause extraction — and finding that most "model failure" was measurement failure

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CUAD V1

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N=40

Public data only (CUAD v1, CC BY 4.0) · Code and full write-up linked

THE INCITING INCIDENT

The number that was wrong

First run: **80%** / **77.5%** / **60%** across governing law, notice period, change of control.

Thirty-three of 120 outputs marked wrong.

I was about to report that change-of-control detection was unreliable.

Instead I read all 33 by hand.

What the numbers said

33

Outputs marked wrong

60%

Change of control accuracy

What the failures actually were

26 of 33 failures were my measurement apparatus.

Real accuracy on change of control: **82.5%**, not 60%.

- ❑ Only 7 of 33 failures were actual model errors.
The eval was the problem.

What the failure actually was	Count
Grading rule never specified (null vs. False)	13
Retrieval bug — excerpt dropped the clause	6
Benchmark label was wrong	5
Grader unit convention (360 vs. 365 days)	2
Model misread the contract	7
Total	33

Setup

Corpus & Fields

Corpus. CUAD v1 — 510 attorney-annotated public contracts. 396 over 15k characters; 91 carry a notice-period annotation; 40 sampled at random from those.

Fields, chosen for contrasting grading properties:

- Governing law — string normalization, 15 distinct values
- Notice period — unit conversion, 15 distinct values
- Change of control — boolean judgment, 25 No / 15 Yes

Model & Retrieval

Model. gemini-3.1-flash-lite, temp 0, Sept 2026. ~744k tokens. Total cost \$0.00.

Retrieval. Keyword-anchored excerpts, median 24% of each document.

510

Contracts in CUAD v1

40

Sampled for this study

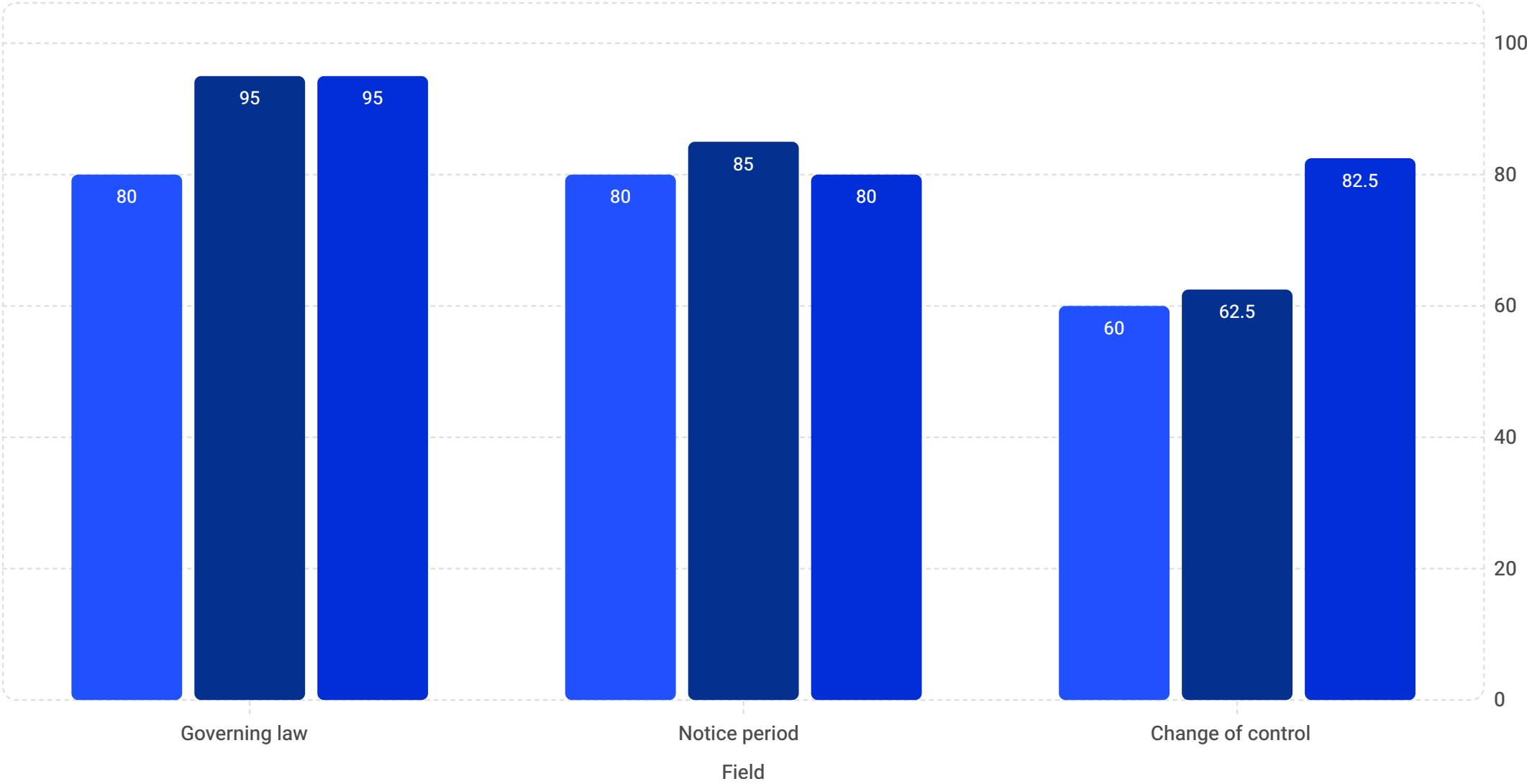
\$0.00

Total inference cost

Three arms, one variable at a time

A → B: retrieval strategy only · B → C: prompt only

Field	A (Baseline)	B (Retrieval fix)	C (Prompt fix)
Governing law	80.0%	95.0%	95.0%
Notice period	80.0%	85.0%	80.0%
Change of control	60.0%	62.5%	82.5%



INTERVENTION ANALYSIS

What each intervention actually did

Interventions are field-specific and non-additive.

Field	Retrieval fix	Prompt fix
Governing law	+15.0 pts (6 ↑ 0 ↓, p=.031)	0.0
Notice period	+5.0 (n.s.)	-5.0 (0 ↑ 2 ↓, n.s.)
Change of control	+2.5 (n.s.)	+20.0 pts (11 ↑ 3 ↓, p=.057)

⚠ The prompt fix regressed a field it wasn't aimed at. A blended accuracy across all three would have shown both changes as modest uniform wins — and hidden the regression completely.

Only 2 of 6 effects clear significance at n=40.

The retrieval bug, visible in the size distribution

Every governing-law failure was a **blank** — never a wrong jurisdiction. That's a retrieval signature, not a comprehension one.

My excerpter filled its budget in document order and stopped.
Governing law lives in the closing "Miscellaneous" section.

133K

Median chars, full corpus

46K

Median chars, sample

Document	Size	Coverage A → B
Upjohn	225,094	12% → 17%
OFG Bancorp	159,336	17% → 25%
Pacira	145,168	19% → 27%
Moelis	122,054	22% → 33%
iPayment	96,662	31% → 40%

Four ways the instrument lied

1

Sampling destroyed the dataset

Shortest-30 for cost reasons — shortest also means simplest. 5 of 30 notice periods, 90/10 class split. Random sampling: 40/40 and 25/15.

2

A metric a constant would have won

At 90/10, always answering "No" scores 90%. One configuration scored 60% overall while catching **0%** of the positive class.

3

An unwritten grading rule

Does abstention count as correctly reporting absence? Never answered, so the grader answered for me. Thirteen failures — the largest category.

4

A grader that wasn't deterministic

The jurisdiction normalizer iterated a Python set. Iteration order varies between processes, so two arms were scored against **different answer keys**.

⊗ Plus: **5 of 40 benchmark labels are wrong** — a ~12% error rate that caps achievable accuracy.

The abstention trade

Forcing a verdict took nulls from **13** → **0** and accuracy from **60%** → **82.5%**.

It also turned three correct answers into confident wrong ones.

Final error profile on change of control

7

False positives

0

False negatives

Right asymmetry for covenant screening — an analyst dismisses a false flag in seconds; a missed provision reaches a credit memo.

- ❑ But I deleted the signal that says "this one needs a human." That should be a deliberate trade, not a side effect of a prompt edit.

What I'd ship

Ship — Governing law extraction

95%, both residual failures are grader artifacts.

Ship as screening aid — Change of control

"Possible CoC language, review §8.2" with the source passage. Never as a populated field.

Don't ship — Notice period

80%, near a ceiling set by label noise, no intervention moved it, and the failure mode is silent.

Restore abstention and route it

Three states: confident answer, abstention with candidate passage, flagged for review.

Regression suite

These 40 documents on every prompt or model change. A 5-point drop in positive-class recall blocks release regardless of aggregate accuracy.

The production metric is analyst override rate

Not benchmark accuracy. One measures agreement with annotators; the other measures whether the people doing the work trust the output.

Limitations, and what I'd do next

Known limitations

n=40 → ±10 point resolution

Only 2 of 6 effects clear it. Full eligible pool is 91.

Conditioned sample

Drawn from contracts that have a notice-period annotation. Not representative.

No inter-annotator agreement measured

I attempted a ceiling estimate, didn't complete it to a standard I trust, and discarded it rather than report it. Largest open gap.

One model, one provider, one date

Results are not portable across model versions.

What comes next

Inter-annotator agreement before the eval.

If two qualified readers don't agree on the right answer, no accuracy number computed against those labels means what it appears to mean.

CLOSING

Optional closing

Most of this harness was AI-generated. The set iteration, the budget truncation, the null handling — I didn't write those lines. I reviewed their outputs and found the defects.

That's the job: validating a system you didn't hand-write, catching failures the system reports as successes.

Resources

Repo: [\[link\]](#) · **Full write-up:** [\[link\]](#)

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No proprietary contracts